

Respiratory: Alveolar ventilation and Dead space 1 of 2

Last updated March 17, 2026  20 min read  CC BY-SA 4.0

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Learning Objectives (5)

After completing this brick, you will be able to:

- 1 Explain the significance of alveolar partial pressures in alveolar gas exchange.
- 2 Describe the importance of alveolar oxygen (P_{AO_2}) and carbon dioxide (P_{ACO_2}) on gas exchange.
- 3 Compare and contrast minute ventilation and alveolar ventilation.
- 4 Explain the impact of the different types of dead space on ventilation.
- 5 Use the Bohr equation to estimate the volume of dead space.



CASE CONNECTION

A 55-year-old male, JB, presents to the emergency room with progressive shortness of breath and fatigue over several months. He is a chronic smoker with a history of COPD. He has a history of COPD. His arterial blood gases show PaO_2 of 60 mmHg and PaCO_2 of 50 mmHg. JM is a first-year medical student shadowing during this visit and notices the abnormal blood gases and is struggling to understand what determines these values and their meaning.

What do JB's blood gas values indicate? How do ventilatory factors contribute to these findings? Consider your answers as you read, and we'll revisit JB at the end of the brick.

[GO TO CONCLUSION](#) ↓

Introduction

This section covers the movement of oxygen and carbon dioxide into and out of the alveoli. Oxygen (O_2) is higher in inspired air than in the alveoli; and its levels depend on the pressure gradient from air to alveolus, ventilation to deliver fresh gas, removal from the alveolus into pulmonary capillary blood. CO_2 is simpler. It is produced by tissue metabolism and moves down its gradient from blood into the alveoli and is exhaled. Thus, blood CO_2 levels primarily reflect ventilation—tidal volume, respiratory rate, and the proportion of dead space.

What Is the Significance of Alveolar Partial Pressures in Alveolar Gas Exchange (LO1)?

Gas exchange in the lungs is driven by **partial pressure gradients**, not by gas concentrations. Oxygen diffuses from alveoli into blood because its partial pressure is higher in the alveoli, while carbon dioxide diffuses from blood into alveoli due to its higher partial pressure in the blood. Understanding partial pressures helps explain how and why gas exchange occurs, and how diseases can disrupt this process.

What is partial pressure (LO1)?

The partial pressure of a gas is the share of total pressure it contributes in a mixture. It is calculated as:

Partial Pressure = (Fractional Gas Concentration) x (Total Atmospheric Pressure).

At sea level, atmospheric pressure is approximately 760 mmHg. Oxygen makes up about 21% of dry air, so $PO_2 = 0.21 \times 760 \text{ mmHg} = 159.6 \text{ mmHg}$. Oxygen partial pressure then decreases stepwise as it moves through the respiratory system and body ([Figure 1](#)).

Alveolar oxygen (P_AO_2) and the Oxygen cascade (LO1 & LO2)

The oxygen cascade describes the stepwise decline in the partial pressures of oxygen (PO_2) in different regions of the body: from inspired air to the mitochondria ([Figure 1](#)). The alveolar oxygen (P_AO_2 , "A" for alveoli) is a key

determinant of respiratory function. It represents the oxygen available to diffuse into the blood and serves as an important clinical marker of gas exchange efficiency.

In the cascade, everything upstream — humidification, CO_2 dilution — determines what $\text{P}_{\text{A}}\text{O}_2$ you achieve, and everything downstream depends on having sufficient $\text{P}_{\text{A}}\text{O}_2$ to maintain the diffusion gradient into blood and onward to tissues.

CLINICAL REASONING

$P_{A}O_2$ influences arterial oxygenation by setting the maximum oxygen available for diffusion into blood. A decrease in $P_{A}O_2$ — due to low inspired oxygen, hypoventilation, or high altitude — reduces oxygen uptake and can cause hypoxemia.

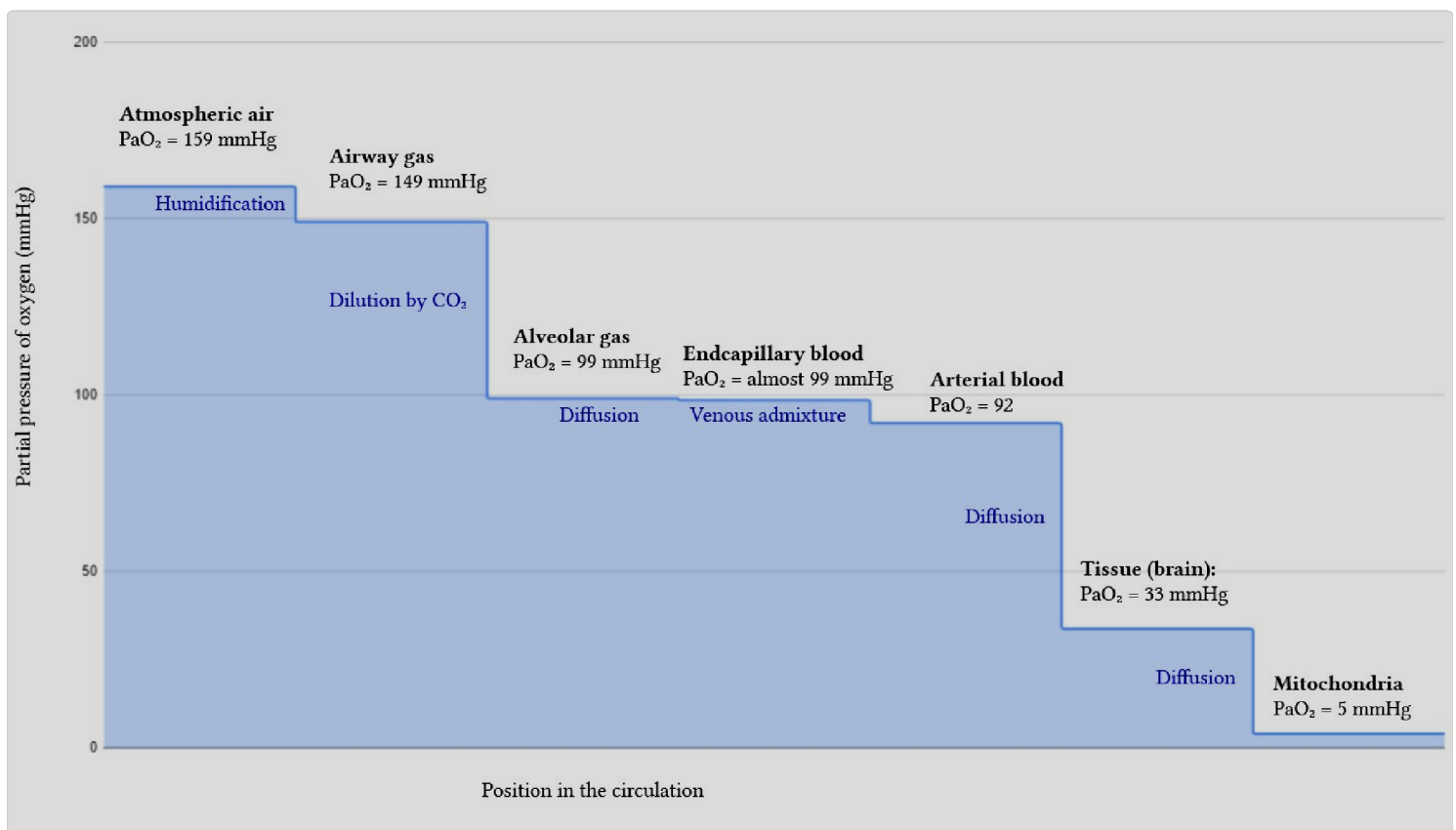


Figure 1. Changes in partial pressures of oxygen from the atmosphere to tissue.

- **Humidification (~10 mmHg reduction: 159 → 149 mmHg).** As dry air enters the upper airways, water vapor at body temperature exerts 47 mmHg of pressure, reducing the share available to other gases. This is obligatory and unavoidable, and always occurs.

- **CO₂ dilution (~50 mmHg reduction: 149 → ~100 mmHg).** Carbon dioxide diffusing from pulmonary capillary blood into the alveolus displaces oxygen. The Alveolar Gas Equation (described in more detail below) captures this relationship precisely, which is why hypoventilation (rising P_aCO₂) directly reduces alveolar oxygen even when inspired oxygen is normal.
- **P_AO₂ is the pivotal step (minimal losses: ~100 → ~99 mmHg in endcapillary blood).** Oxygen diffuses down its partial pressure gradient from the alveoli (P_AO₂ ≈ 100 mmHg) into deoxygenated blood arriving from the pulmonary arteries (P_aO₂ ≈ 40 mmHg, not shown in [Figure 1](#)). The alveolar-capillary barrier is extremely thin and the surface area massive. Thus, under normal conditions blood fully equilibrates during its transit through the capillary surrounding the alveoli (~99 mmHg), with minimal losses in health ([Figure 1](#)). Importantly, once oxygen enters the blood, it is the partial pressure of dissolved oxygen (P_aO₂) — not hemoglobin-bound oxygen — that drives diffusion onward to the tissues.
- **Blood mixing (~7 mmHg loss: 99 → 92 mmHg). Bronchial circulation** (blood supplying the upper respiratory tissues) and **thebesian veins** (blood supplying the heart tissues) return deoxygenated blood directly into the arterial supply, diluting the PO₂. This is the physiological basis of the normal A-a gradient — and why arterial partial pressure of oxygen is always slightly lower than alveolar levels (P_aO₂ vs P_AO₂), even in a perfectly healthy lung.
- **Tissue diffusion (the steepest drop: 92 → 33 mmHg in brain, down to ~5 mmHg at mitochondria).** This is where oxygen is actually consumed. The gradient must be steep enough to drive O₂ from capillary blood through interstitial fluid and into cells. Metabolically active

tissues maintain very low intracellular PO_2 , which sustains the driving pressure for continuous delivery.

Alveolar Gas Equation– Oxygen in the Alveoli

The **Alveolar Gas Equation** (Figure 2) helps clinicians understand how the below factors affect alveolar oxygen levels:

- Oxygen removal by diffusion across the alveolar barrier into blood
- Oxygen replenishment through ventilation (breathing)
- Carbon dioxide production

For example, at **high altitude** where ambient pressure (P_{atm}) is reduced, $P_{\text{A}}\text{O}_2$ decreases proportionally. Conversely, the **supplemental oxygen** therapy increases the fraction of oxygen in the inhaled gas ($F_{\text{i}}\text{O}_2$, typically from 0.21 to 0.5-1.0), which raises $P_{\text{A}}\text{O}_2$ according to the alveolar gas equation. This is why oxygen therapy is beneficial for patients with **hypoxemia**. Similarly, in hypoventilation states where $P_{\text{A}}\text{CO}_2$ rises, $P_{\text{A}}\text{O}_2$ falls even if inspired oxygen remains constant.

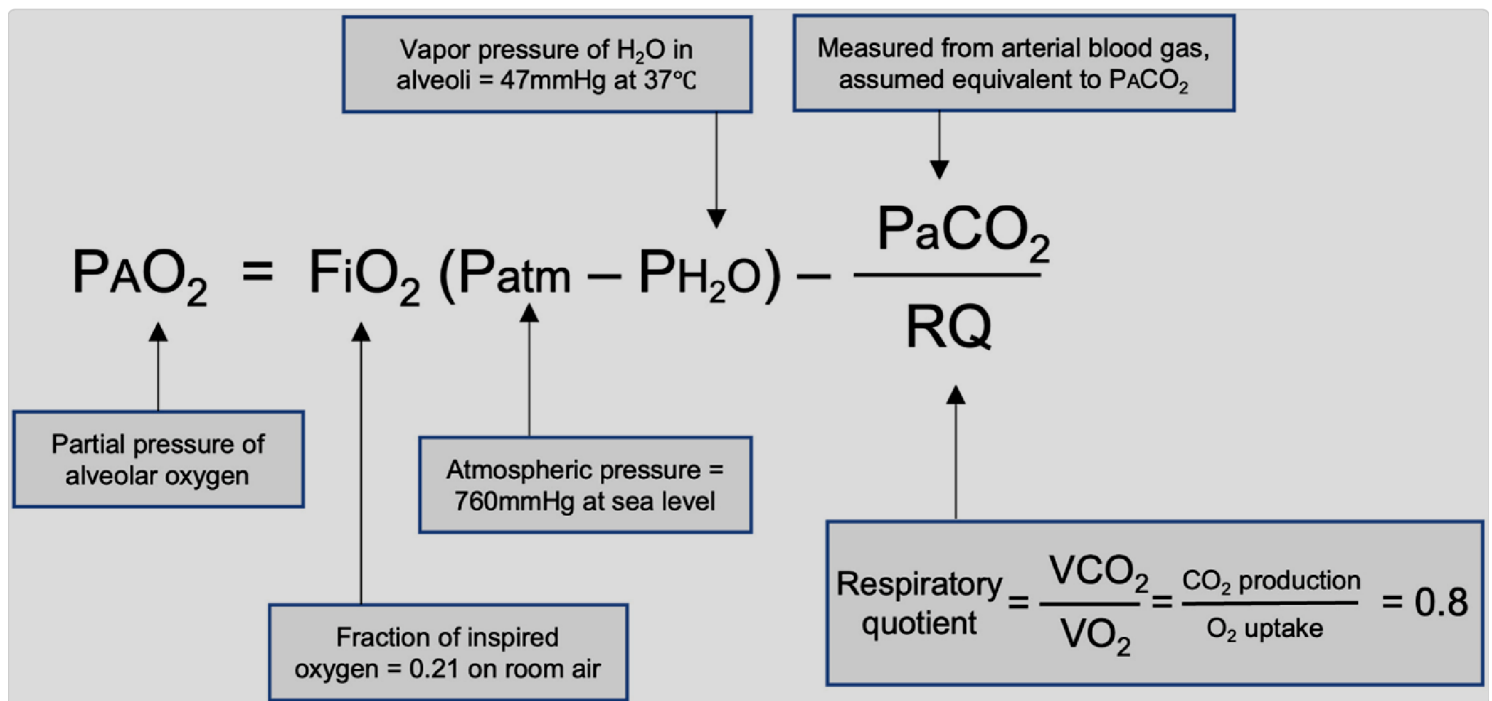


Figure 2. Alveolar Gas Equation

P_iO_2 is the *partial pressure of inspired oxygen* in the upper airways after humidification. When it reaches the alveoli, CO_2 moves out of your blood and into the alveoli, "crowding out" some of the oxygen. This is why we subtract the P_aCO_2/RQ from the P_iO_2 in the alveolar gas equation. At sea level the P_iO_2 works out to 150 mmHg ($0.21 \times (760 \text{ mmHg} - 47 \text{ mmHg}) \approx 150 \text{ mmHg}$), allowing the equation to be simplified:

- **R:** The respiratory quotient (usually 0.8) represents the ratio of carbon dioxide production to oxygen consumption during cellular metabolism. Note that an all-carb diet, strenuous exercise, or severe illness all increase "R" because the body makes more CO_2 . Low- or no-carb diets will reduce the ratio. If a clinical vignette does not define an R quotient, assume the value to be 0.8.

ACTIVE LEARNING QUESTION

Using the Alveolar Gas Equation, what happens to P_AO_2 when a patient with normal lungs ascends to high altitude ($P_iO_2 = 100 \text{ mmHg}$) and maintains a P_aCO_2 of 35 mmHg?

- A. PAO_2 increases

- B. PAO_2 remains unchanged
- C. PAO_2 decreases to ~40 mmHg
- D. PAO_2 decreases to ~56 mmHg
- E. PAO_2 decreases to ~65 mmHg

Explanation (requires correct answer)



CLINICAL REASONING

Changing the partial pressure of inspired oxygen (PiO_2):

Pressure of inspired O_2 : The fraction of inspired oxygen (FiO_2) is always ~21% in ambient air, even at high altitude. What changes is air pressure: at higher elevations, total pressure falls, so PiO_2 decreases, reducing oxygen availability.

Fraction (Percentage) of inspired O_2 : F_iO_2 can be altered clinically. For example, a patient prescribed $F_iO_2 = 1.0$ therapy will receive 100% O_2 when intubated because it is a closed system. Using an open system which allows leakage of oxygen, a standard facemask will allow F_iO_2 to reach only 40 - 60% despite a full flow of oxygen (10 - 15 L/min). Nasal cannula only allow oxygen to flow at a maximum of 6 L/min, limiting the F_iO_2 to 44%.



CLINICAL CORRELATION

Hyperbaric oxygen (HBO) therapy: Used for conditions like decompression sickness and carbon monoxide poisoning, this therapy delivers up to 100% oxygen at 2–3 times normal atmospheric pressure in a special chamber, increasing the partial pressure of O_2 . Since hemoglobin is already near 100% saturated with the use of supplemental oxygen, HBO increases the total O_2 content of blood only by increasing the *dissolved* content in the plasma. In decompression sickness, the very low alveolar nitrogen partial pressure (near 0%) favors the accelerated diffusion of excess nitrogen bubbles from the blood into the alveoli where it is exhaled and removed from the body. In carbon monoxide poisoning, the high O_2 content displaces CO bound to hemoglobin and CO is then removed through respiration.

Partial pressure gradients

Alveolar carbon dioxide ($P_A\text{CO}_2$) (LO_2)

Unlike oxygen, which moves from the atmosphere into the blood, CO_2 flows in the opposite direction. Because atmospheric CO_2 is only ~0.04%, nearly all exhaled CO_2 is produced by tissue metabolism and diffuses from pulmonary capillary blood ($P_a\text{CO}_2 \approx 45 \text{ mmHg}$) into the alveoli ($P_A\text{CO}_2 \approx 40 \text{ mmHg}$). As a result, alveolar CO_2 reflects the balance between metabolic CO_2 production and its removal by ventilation.

Next, since CO_2 equilibrates so quickly across the alveolar membrane that $P_A\text{CO}_2$ and $P_a\text{CO}_2$ are nearly identical under normal conditions. For this reason, **$P_a\text{CO}_2$ serves as a direct but inverse marker of ventilation**: it rises with hypoventilation and falls with hyperventilation (See [Table 1](#), and equation - *which explains the intuitive relationship*).

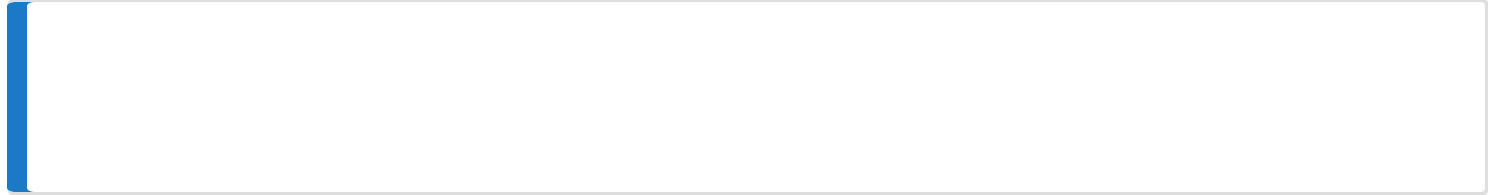
Table 1. Alveolar partial pressures in different clinical states

Clinical State	P _A O ₂	P _A CO ₂	Mechanism
Hypoventilation	Decreased	Increased	Inadequate ventilation relative to CO ₂ production
Hyperventilation	Increased	Decreased	Excessive ventilation relative to CO ₂ production
High Altitude	Decreased	Decreased	Lower P _i O ₂ ; compensatory hyperventilation
Chronic Obstructive Pulmonary Disease (COPD)	Decreased	Often increased	Ventilation-perfusion mismatch; increased dead space

Hypoventilation, such as in COPD or respiratory muscle weakness.

Minute ventilation vs alveolar ventilation (LO3)

Minute ventilation ($\dot{V}_E$) is the total volume of air moved in and out of the lungs per minute. This is shown in the equation below where the volume of air moved is tidal volume (V_t) and the respiration rate as RR:



With a tidal volume of 500 mL and a rate of 12 breaths/min, minute ventilation is 6,000 mL/min, or 6 L/min.

However, not all of this air in the respiratory system participates in gas exchange. Each breath includes a portion that never reaches the alveoli, or participates in gas exchange—this is **dead space**. We need a second measurement to understand how much air is actually exchanged per breath between the respiratory spaces of the lung and the external environment:

Alveolar ventilation ($\dot{V}_A$) is the volume of exchanged air reaching the alveoli per minute:

Using a dead space of 150 mL: $\dot{V}_A = (500 - 150) \times 12 = 4,200$ mL/min. Of the 6 L/min of total minute ventilation, only 4.2 L is participating in gas exchange. The remaining 1.8 L/min is wasted as dead space ventilation.

wasted ventilation.

alveolar ventilation. Multiplied by the total breaths, this is a lot of
The deadspace gets subtracted from each breath to calculate

In summary, we described ventilation by the exchange of CO₂ as eupnea (normal), hyperventilation, or hypoventilation. P_ACO₂ and P_aCO₂ are affected by the ventilation rate, tidal volume, and dead space- where alveolar and minute ventilation calculations are important. Last, P_aCO₂ is also affected by the CO₂ produced by the body through a higher metabolic rate which increases arterial, and thus alveolar CO₂ levels.

Why Breathing Pattern Matters – getting gases to the alveoli

Table 2 illustrates a critical concept: three different breathing patterns can produce the same minute ventilation (6000 mL/min) but dramatically different alveolar ventilation.

Notice the first row: if tidal volume is 200 mL, only 50 mL of fresh air reaches the alveoli — alveolar ventilation plummets. The CO₂ would skyrocket to lethal levels. Conversely, slow, deep breathing (last row) maximizes alveolar ventilation because a smaller proportion of each breath is wasted.

Table 2. Effects of frequency and depth of breathing (Vt) - dead space is 150 mL for all 3 subjects

Breathing pattern	Tidal Volume (Vt) (mL)	Frequency (breaths/min)	Minute Ventilation (mL/min)	Air "trapped" in Dead Space (mL/min)	Alveolar Ventilation (mL/min)
Rapid, shallow	200	30	6000	150 × 30 = 4,500	1,500
Normal	500	12	6000	150×12 = 1,800	4,200

Breathing pattern	Tidal Volume (Vt) (mL)	Frequency (breaths/min)	Minute Ventilation (mL/min)	Air "trapped" in Dead Space (mL/min)	Alveolar Ventilation (mL/min)
Slow, deep	1000	6	6000	$150 \times 6 = 900$	5,100

This explains why patients in respiratory distress who breathe rapidly and shallowly may develop hypercapnia and hypoxemia despite increased minute ventilation — their alveolar ventilation is inadequate. This concept is especially important in mechanical ventilation, where optimizing tidal volume and rate improves alveolar ventilation while minimizing ventilator-induced lung injury.

ACTIVE LEARNING QUESTION

A patient has a tidal volume of 400 mL, respiratory rate of 15 breaths/min, and dead space of 150 mL. What is the alveolar ventilation?

- A. 6 L/min
- B. 2.25 L/min
- C. 3.75 L/min
- D. 4.5 L/min
- E. 5.25 L/min

Explanation (requires correct answer)

Different types of dead space (LO4)

Now let's look more closely at dead space itself. **Dead space** is the portion of each breath that does not participate in gas exchange. It represents "wasted" ventilation and is a major determinant of respiratory efficiency.

Understanding the types of dead space explains why patients with certain conditions develop hypoxemia and hypercapnia despite seemingly adequate breathing.

To visualize this, consider what happens during exhalation (Figure 3). The first air out comes from the conducting airways and contains almost no CO₂ (atmospheric CO₂ is only 0.04%). Then CO₂-rich air follows, coming from deeper within the lungs, the alveoli. If we could completely empty the lungs with each breath, expired partial pressure of carbon dioxide (P_ECO₂) would

equal $P_A\text{CO}_2$ — but the lungs never fully empty. Thus there are regions of the lungs that do not aid in the removal of CO_2 , or bringing O_2 to the blood.

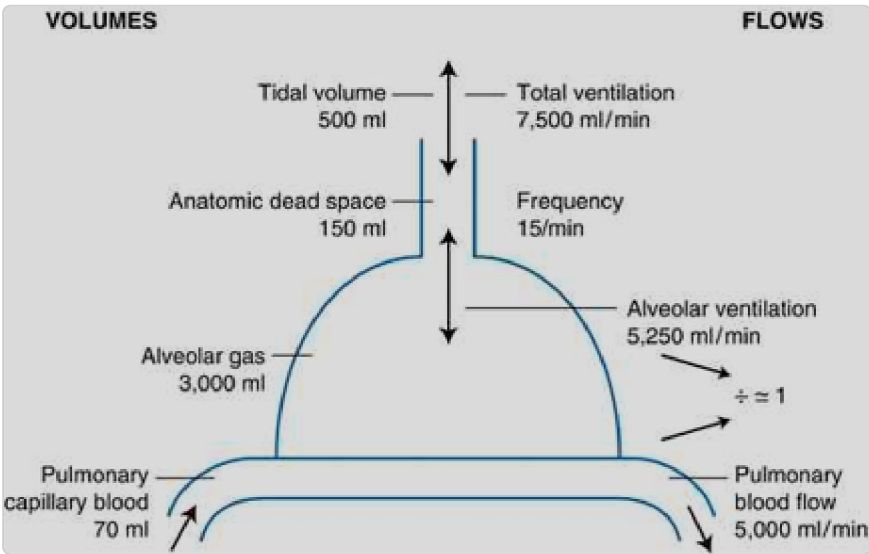


Figure 3. Diagram of lung showing ventilation and blood volumes and flows

There are three main types of dead space, with examples given in Table 3:

- **Anatomic Dead Space:** The volume of the conducting airways (nose, pharynx, trachea, bronchi) where no gas exchange occurs—approximately 150 mL in adults, or ~1 mL per pound of ideal body weight.
- **Alveolar Dead Space (also called pathological dead space):** Refers to alveoli that are ventilated but not perfused (or poorly perfused), so no effective gas exchange occurs. In healthy individuals, this is minimal.
- **Physiologic Dead Space:** The sum of anatomical and alveolar dead space— the total "wasted" ventilation per breath.

Table 3. Clinical examples of dead space

Condition	Type of Dead Space Affected	Mechanism
Emphysema	Physiological	Destruction of alveolar walls and capillary beds
Pulmonary Embolism	Alveolar	Blocked pulmonary vessels creating ventilated but unperfused regions
Mechanical Ventilation	Anatomical	Endotracheal tubes and ventilator circuits increase conducting airway volume
Acute Respiratory Distress Syndrome (ARDS)	Physiological	Alveolar collapse and microvascular injury creating V/Q mismatch



perfused alveoli.

physiologic dead space includes anatomic dead space plus non-
Anatomic dead space is the air in the conducting passages, while

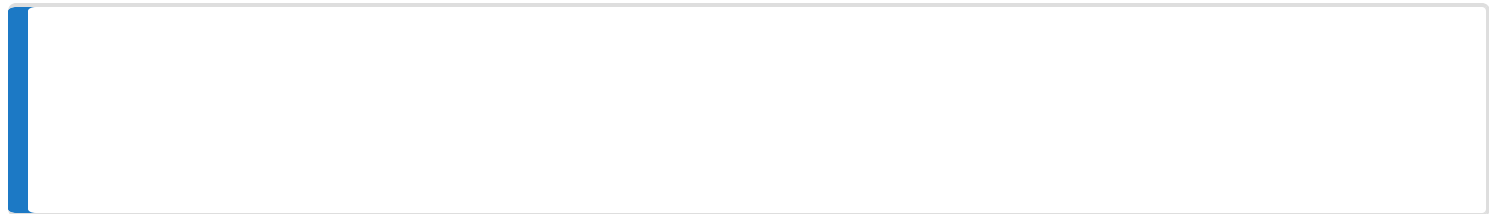


Impact of dead space on ventilation and gas exchange (LO4)

When dead space increases, less fresh air reaches the alveoli, leading to reduced O₂ delivery to blood (hypoxemia) and inefficient CO₂ removal (hypercapnia, which can cause respiratory acidosis). To compensate, the body increases breathing rate and depth, raising the work of breathing and promoting fatigue. In mechanically ventilated patients, the added dead space from tubing is factored into tidal volume calculations.

How Is the Bohr Equation Used to Estimate Dead Space Volume? (LO5)

The **Bohr equation** estimates physiologic dead space using the difference between arterial and expired CO₂. It provides the ratio of dead space volume (V_D) to tidal volume (V_t):



Where:

- V_D/V_t : The ratio of dead space volume to tidal volume.
- P_aCO_2 : The partial pressure of carbon dioxide in the arterial blood.
- P_ECO_2 : The partial pressure of carbon dioxide in the expired (exhaled) air.

The principle is straightforward: dead space contains no CO_2 at the end of inhalation (it's fresh air), so it dilutes the CO_2 concentration in expired air. If there were no dead space, expired CO_2 (P_ECO_2) would equal alveolar CO_2 (P_ACO_2). Because dead space exists, P_ECO_2 is always diluted by air in the dead space volume and lower than P_ACO_2 . Since alveolar CO_2 cannot be measured directly, arterial CO_2 (P_aCO_2) is used as a proxy. This calculation is valuable in clinical settings to assess ventilation efficiency and diagnose conditions that increase dead space ventilation.



CLINICAL REASONING

How does dead space affect $PECO_2$? In emphysema, destruction of alveolar walls and pulmonary capillaries creates regions that are ventilated but poorly perfused, increasing alveolar dead space. Thus, the increased difference between $PaCO_2$ and $PECO_2$ reflects inefficient ventilation and helps clinicians estimate the degree of dead space using the Bohr equation.

Partial pressure of arterial CO_2 ($P_a\text{CO}_2$)

($P_a\text{CO}_2$): P_{ECO_2} = Partial pressure of expired CO_2 (P_{ECO_2}): $P_a\text{CO}_2 =$
 $T_a\text{CO}_2$ = Tidal volume (T_V): $P_a\text{CO}_2$ = Partial pressure of arterial CO_2

Interpreting dead space calculations (Bohr equation)

The V_D/V_t ratio provides valuable information about ventilation efficiency:

- **Normal V_D/V_t :** In healthy individuals, 20-40% of each breath is dead space and there is efficient gas exchange and effective CO_2 removal.
- **Elevated V_D/V_t :** High values indicate increased dead space relative to tidal volume, and inefficient gas exchange. The reduced exhalation of CO_2 (P_{ECO_2}) increases $P_a\text{CO}_2$, and a buildup of CO_2 in the blood (**hypercapnia**).

The Bohr equation helps clinicians understand the efficiency of ventilation and guides therapeutic interventions. For example, in mechanically ventilated patients, a rising V_d/V_t may signal worsening lung function or inappropriate ventilator settings, while a falling ratio suggests improvement.

(hypoxemia) and increased CO_2 levels (hypercapnia).
It reduces efficient ventilation, leading to reduced oxygenation

CASE CONNECTION

[BACK TO INTRODUCTION](#) 

Thinking back to JB, his arterial blood gases show hypoxemia ($\text{PaO}_2 = 60$ mmHg) and hypercapnia ($\text{PaCO}_2 = 50$ mmHg). The low PaO_2 indicates impaired oxygen uptake, which in COPD can result from reduced ventilation and diffusion limitation due to alveolar destruction. The elevated PaCO_2 reflects inadequate ventilation (obstructive disease) and increased dead space, both characteristic of COPD. Because COPD is obstructive, air becomes trapped, alveoli are damaged, and gas exchange efficiency falls. These findings together confirm impaired gas exchange due to ventilatory defects, consistent with chronic lung disease.

Summary

- Ventilation is the process of supplying oxygen to and removing carbon dioxide from the lungs, which is driven by the partial pressures of these gases.
- Gas partial pressure gradients drive the diffusion or passive movement of gases between the alveoli and the pulmonary capillaries.

- The partial pressure of oxygen (PO_2) varies throughout the respiratory system, with changes from the mouth to the alveoli being described by the oxygen cascade.
- The Alveolar Gas Equation - accurately measures PAO_2 - taking into account oxygen removal by blood, and replenishment through breathing.
- The alveolar ventilation equation - which measured $PACO_2$ - illustrates that $PACO_2$ is markedly altered by alveolar ventilation.
- The partial pressures of carbon dioxide (PCO_2) can be used to calculate ventilation efficiency, and dead space.
- There are different types of dead space (physiologic, anatomical, and alveolar), and increased dead space leads to reduced O_2 uptake and CO_2 excretion.
- Dead space accounts for the differences between total minute ventilation and alveolar ventilation calculations.
- Dead space can be calculated using the Bohr equation, using the partial pressure of carbon dioxide in the arterial blood and the expired air (mnemonic: TaCo PaCo PECO PaCo).

Review Questions

1. What drives the movement of oxygen from the alveoli into the blood and carbon dioxide out of the blood?

- A. Gas concentration in the alveoli
- B. Partial pressure gradients
- C. Ventilation rate
- D. Alveolar surface area

Explanation (requires correct answer)

2. In which disease would you expect to see increased alveolar dead space?

- A. Pulmonary embolism
- B. Acute asthma attack
- C. Bronchitis
- D. Pneumothorax

Explanation (requires correct answer)

3. A 65-year-old male presents with PaO_2 of 55 mmHg and PaCO_2 of 60 mmHg. Which of the following best describes his gas exchange status?

- A. Hyperventilation
- B. Impaired alveolar ventilation
- C. High FiO_2
- D. Normal gas exchange

Explanation (requires correct answer)

A.

B.

- C. 5,250 mL/min
- D. 6,000 mL/min

Explanation (requires correct answer)

5. A 150 lb patient scheduled for abdominal surgery is sent for preoperative evaluation and testing. His chest x-ray is normal, and pulmonary function results on room air show the following: Tidal volume = 600 mL, resp rate = 12 /min; $\text{PaO}_2 = 90$ mmHg; $\text{PaCO}_2 = 40$ mmHg; $\text{PECO}_2 = 28$ mmHg. What is the patient's dead space?

- A. 0.13 mL
- B. 150 mL
- C. 180 mL
- D. 420 mL
- E. 7200 mL

Explanation (requires correct answer)

How would you rate this Brick?



Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

- 01 Explain the significance of alveolar partial pressures in alveolar gas exchange.
- 02 Describe the importance of alveolar oxygen ($P_{A}O_2$) and carbon dioxide ($P_{A}CO_2$) on gas exchange.
- 03 Compare and contrast minute ventilation and alveolar ventilation.

04

Explain the impact of the different types of dead space on ventilation.

05

Use the Bohr equation to estimate the volume of dead space.

Objective Confidence: 0/5